



Central Assessment of Full-Limb X-rays for Frontal Plane Lower Limb Alignment

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1. Overview

1.1 SAS dataset

Name: **flXR_KneeAlign_CookeXX** (XX identifies the time point)

Display label: **flXR knee alignment (Cooke)**

Name: **flXR_KneeAlign_DuryeaXX** (XX identifies the time point)

Display label: **flXR knee alignment (Duryea)**

1.2 Contents of datasets

These datasets contain central measurements of frontal plane lower limb mechanical alignment (hip-knee-ankle angle) from OAI full-limb x-rays. These were performed independently by two separate reading centers.

Measurements in the flXR_KneeAlign_CookeXX datasets (OAI Project 60) were funded by OAI and performed by OASYS Inc. (Dr. Derek Cooke www.oaisysmedical.com) with support of staff at Queen's University, Kingston, Ontario.

Measurements in the flXR_KneeAlign_DuryeaXX datasets (OAI Project 32) performed independently of OAI in the laboratory of Dr. Jeff Duryea at Brigham and Women's Hospital in Boston, MA. <http://www.spl.harvard.edu/pages/People/duryea> and include tibia and femur length also.

Each dataset contains the reading data for a single visit (or time point) from a single vendor. The corresponding longitudinal data from reading projects by these vendors for various time points are in separate datasets. The data from these vendors for the 12-month visit x-ray readings are in datasets ending in "01" (i.e., flXR_KneeAlign_Cooke01 for Dr. Cooke's readings and flXR_KneeAlign_Duryea01 for



Dr. Duryea's readings), while the corresponding data from these vendors for the 24-month visit x-ray readings are in datasets with the same name but ending in "03" (i.e., flXR_KneeAlign_Cooke03 for Dr. Cooke's readings and flXR_KneeAlign_Duryea03 for Dr. Duryea's readings), and so on. (See the "[VisitPrefixDefinitions.pdf](#)" document for a guide to visit numbering.) To compare values of a variable across time points from a given project by the vendor, or to calculate change scores, users will need to merge the datasets for the various time points.

IMPORTANT: All users are strongly encouraged to read the *Overview and Description of Central Image Assessments* document carefully before attempting to use this dataset!

For comprehensive and detailed information about the structure and contents of the central image assessment datasets, general guidelines for defining change between time points and duration of follow-up, and strategies for merging datasets for analysis, please see the "[Overview and Description of Central Image Assessments](#)" document. This document is located in the zip compressed documentation file.

1.3 Condition

- Known data errors: problems/cautions for use are listed by variable in the "Release Comments."
- Dataset strengths/weaknesses:
 - o Data are expected for all participants included in a project sample. If expected data do not exist for a knee, SAS special missing values are assigned to denote why the data were not obtained.
 - o The datasets may include multiple rows of data (records) for a given knee which needs to be taken into account when merging it with other datasets. Please see the "[Overview and Description of Central Image Assessments](#)" document for more information on merging.

1.4 Variables and reading methods

See the dataset documentation files [flXR_KneeAlign_CookeXX_Contents.pdf](#) and [flXR_KneeAlign_DuryeaXX_Contents.pdf](#) in the compressed documentation files for a complete list of all the variables in the datasets, their SAS variable names, descriptive variable labels, and attributes.

Variables assessed for all projects by these vendors include:

- hip-knee-ankle (HKA) angle.

Any additional project-specific variables are described below for each project.

The measurement techniques and variables are similar in the different projects by these vendors, and are outlined below. Any modifications for a specific project are described in the section related to that project, including:

- criteria for selecting participants to include.

In general, frontal plane lower limb alignment is measured from standing bilateral radiographs of the lower limbs in their entirety, with knees fully extended (called 'full-limb' radiographs in OAI). Positional reliability is provided for by aligning the 'knee flexion plane' ahead. Frontal alignment may be defined as the angle between the mechanical axes of the femur and tibia. This angle has been termed the hip-knee-ankle (HKA) angle or the mechanical axis angle.

These publications give more details about the methods and measurement performance for data in the flXR_KneeAlign_CookeXX datasets (project 60):

- Cooke TD, Sled EA, Scudamore RA. Frontal plane knee alignment: a call for standardized measurement. *J Rheumatol.* 2007; 34:1796-801. PMID: 17787049.
- Cooke TD, Harrison L, Khan B, Scudamore A, Chaudhary MA. Analysis of limb alignment in the pathogenesis of osteoarthritis: a comparison of Saudi Arabian and Canadian cases. *Rheumatol Int.* 2002; 22(4): 60-164. <http://dx.doi.org/10.1007/s00296-002-0218-7> PMID: 12172956.
- Sled E, Sheehy L, Felson DT, Costigan, PA, Lam M, Cooke TDV, Reliability of lower limb alignment measures using an established landmark-based method with customized computer software program. *Rheumatol Int* 2011; 31(1): 71-77. <http://dx.doi.org/10.1007/s00296-009-1236-5> PMID: 19882339 PMCID PMC3048894.

From the anatomical and functional perspective, the orientation of the femur and tibia at the knee is best described in terms of the bones' mechanical axes. The orientation of these axes reflects alignment in stance, which may be neutral, varus (bowlegged), or valgus (knock-kneed) (Figure 1).

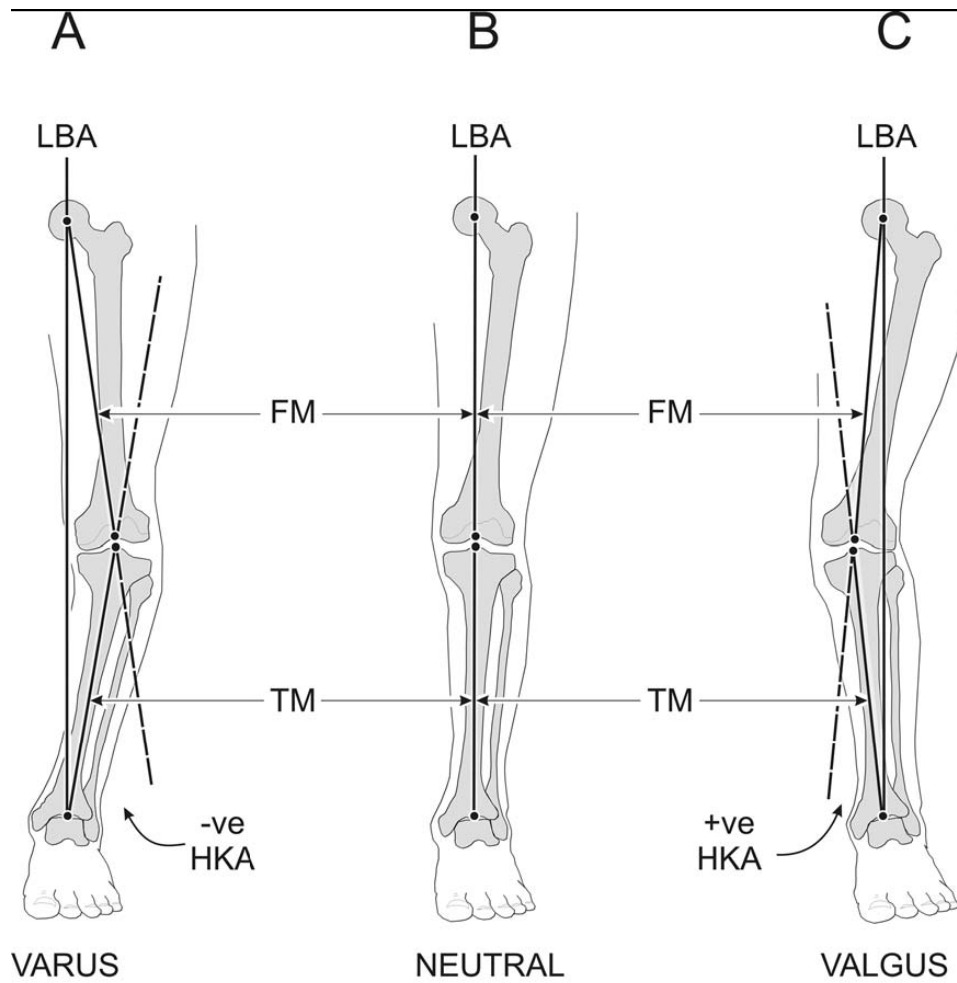


Figure 1. Common frontal plane lower limb alignment patterns. A. Varus alignment: knee center is lateral to the LBA (HKA is negative). B. Neutral alignment: knee center is located on the LBA (HKA = 0°); femoral and tibial mechanical axes are colinear. C. Valgus alignment: knee center is medial to the LBA (HKA is positive). LBA: load-bearing axis, HKA: hip-knee-ankle angle, FM: femoral mechanical axis, TM: tibial mechanical axis. (From reference 1).



The mechanical axis of the femur (FM) is located as a line from the center of the femoral head running distally to the mid-condylar point between the cruciate ligaments. In the case of the tibia, the mechanical

axis (TM) is a line from the center of the tibial plateau (interspinous intercruciate midpoint) extending distally to the center of the tibial plafond. The angle between these two axes is the hip-knee-ankle (HKA) angle. In the neutrally-aligned limb the HKA angle approaches 180° . At this point FM and TM are co-linear, pass through the knee center, and are coincident with the load-bearing axis (LBA) which is the line of ground reaction force passing from the ankle to the hip (Figure 1B). In *varus* the knee center is lateral to the LBA (Figure 1A), whereas in *valgus* the knee center is displaced medially (Figure 1C). As a convention the HKA is expressed as its angular deviation from 180° (i.e. HLA = 0° in neutral alignment). Varus deviations are negative and valgus ones are positive. The choice of varus as a negative value and valgus as positive is based on the general observations of a more serious problem of loading and damage in the varus knee.

Performance characteristics of measurements of HKA angle performed by OAISYS Inc. (Project 60) are shown in figure 2.

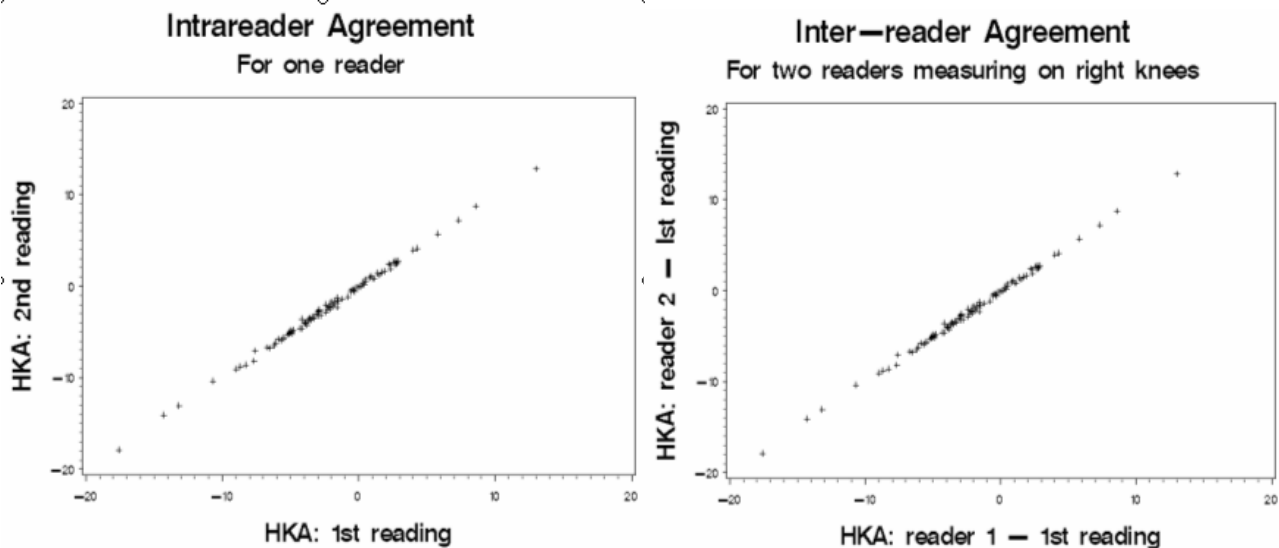


Figure 2. Intrareader and Inter-reader reliability of HKA angle from repeated measurements by OAISYS Inc. made on the same digital images by 3 readers who independently identified and marked anatomic landmarks. Angular dimensions were automatically derived from these landmarks by a customized computer software program. Intrareader intra-class correlation coefficient for 3 readers was 0.99 (n=200 limbs). Inter-reader agreement for three readers was 0.99 (n=200 limbs). (Reference 6)

There are no comparable publications of reliability and measurement methods for HKA measurements performed by Dr. Jeff Duryea's laboratory (Project 32), but there is good agreement between the measurements from Project 32 with those from Project 60 (Dr. Derek Cooke's laboratory).



2. Methods specific to Project 60

2.1 Image type:

Bilateral full-limb radiograph. See the “*Radiographic (X-ray) Manual*” operation manuals for the acquisition protocol.

2.2 Time points:

12-month, 24-month, 36-month, and 48-month visits.

2.3 Measurement methods:

Prior to performing measurements, each reader's capability was compared to standards of 3 experienced staff. Images were analyzed as *.tiff or *.bmp files using the Horizon Surveyor (version 1.5, OASYS Inc.) custom software program to define the femoral and tibial bone landmarks. Magnification and enhancement of brightness and contrast of the image were employed as needed. The reader was prompted to identify each landmark in a defined sequence and the data for the landmark's X-Y coordinates were collected in a customized data collection template. Readers were trained to identify images in which there were rotational positional errors (Reference 5). Before being submitted, data were reviewed by 'expert' readers, including spot checks of repeat measurements to compare results. Images for evaluation as *.tiff or *.bmp files in batches of 50 were uploaded to a secure ftp site from which readers, trained by OASYS staff, downloaded the images for analysis.

2.4 Variables:

For this project:

- HKA angle of right and left lower limbs (VxxHKANGLE).

2.5 Sample:

Participants in the progression and incidence subcohorts of OAI were eligible for acquisition of full-limb radiographs, and anyone who had a full-limb radiograph which was readable had HKA measured. Initially, full-limb radiographs were only to be acquired in the progression subcohort and that acquisition began at the 12-month follow-up visit; but for progression subcohort participants in whom no 12-month full-limb radiograph was acquired, attempts were made at later visits (usually the 24-month visit) to acquire one. Funding for acquisition of full-limb radiographs in the incidence subcohort was obtained later and their full-limb radiographs were generally acquired at the 24-month visit; but again, later visit acquisitions were attempted when no suitable x-ray was obtained at the 24-month visit. Only one full-limb radiograph was acquired per person unless the initial one was of poor quality, in which case a later visit image may also have been obtained. The numbers of participants with HKA measurements obtained at each visit (by subcohort) is listed below.

Table 1. Showing the numbers of participants in Progression and Incidence subcohorts with HKA measurements obtained at each visit (Project 60)

Visit at which HKA was measured	Progression Subcohort	Incidence Subcohort	Total
12-month visit (V01)	994	243	1,237
24-month visit (V03)	131	1,144	1,275
36-month visit (V05)	50	861	911
48-month visit (V06)	9	123	132



3. Methods specific to Project 32

3.1 Image type:

Bilateral full-limb radiograph. See the “*Radiographic (X-ray) Manual*” operation manual for the acquisition protocol.

3.2 Time points:

12-month, 24-month, 36-month, and 48-month visits.

3.3 Measurement methods:

OAI publically released DICOM images were loaded into custom software and HKA was calculated from 3 points defined on the full-limb radiographs. To define the location of the femoral head, the reader placed three seed points approximately evenly distributed on the margins of the femoral head. The first point was defined as the center of the unique circle that passed through these points. A second point was placed in the approximate center of the region defined by the tibial spines and femoral notch. For the third point, the reader first placed landmarks on the medial and lateral edges of the talus at the tibio-talar joint. A vertical line bisecting these point was displayed and the reader marked the third HKA point where this line intersected the proximal margin of the talus. The HKA was then defined as the angle subtended between the line through the center of the femoral head and the center of the knee with the line through the center of the tibio-talar joint. As well as measurements of HKA, the length of the femur and the length of the tibia are provided. These lengths are the lengths of FM (for femur) and TM (for tibia) shown in Figure 1.

3.4 Variables:

For this project:

- HKA angle of right and left lower limb (VxxHKANGJD).

and new variables since the last release:

- Femur Length (VxxFEMLEN).
- Tibia Length (VxxTIBLEN).

3.5 Sample:

Participants in the progression and incidence subcohorts of OAI were eligible for acquisition of full-limb radiographs, and anyone who had a full-limb radiograph which was readable had HKA measured. Initially, full-limb radiographs were only to be acquired in the progression subcohort and that acquisition began at the 12-month follow-up visit; but for progression subcohort participants in whom no 12-month full-limb radiograph was acquired, attempts were made at later visits (usually the 24-month visit) to acquire one. Funding for acquisition of full-limb radiographs in the incidence subcohort was obtained later and their full-limb radiographs were generally acquired at the 24-month visit; but again, later visit acquisitions were attempted when no suitable x-ray was obtained at the 24-month visit. Only one full-limb radiograph was acquired per person unless the initial one was of poor quality, in which case a later visit image may also have been obtained. The numbers of participants with HKA measurements obtained at each visit (by subcohort) is listed below.

Table 2. Showing the numbers of participants in Progression and Incidence subcohorts with HKA measurements obtained at each visit (Project 32)

Visit at which HKA was measured	Progression Subcohort	Incidence Subcohort	Total
12-month visit (V01)	1,037	395	1,432
24-month visit (V03)	130	1,131	1,261
36-month visit (V05)	50	864	914
48-month visit (V06)	15	161	176



4. References

1. Cooke TD, Sled EA, Scudamore RA. Frontal plane knee alignment: a call for standardized measurement. J Rheumatol. 2007; 34:1796-801. PMID: 17787049.
2. Cooke TD, Harrison L, Khan B, Scudamore A, Chaudhary MA. Analysis of limb alignment in the pathogenesis of osteoarthritis: a comparison of Saudi Arabian and Canadian cases. Rheumatol Int. 2002; 22(4):160-164. <http://dx.doi.org/10.1007/s00296-002-0218-7> PMID: 12172956.
3. Cooke TD, Scudamore RA, Bryant JT, Sorbie C, Siu D, Fisher B. A quantitative approach to radiography of the lower limb. Principles and applications. J Bone Joint Surg Br. 1991; 73:715-20. PMID: 1894656
4. Cooke TDV and Sled E: Optimizing Limb Position for Measuring Knee Anatomical Axis Alignment from Standing Knee Radiographs. J.Rheum. 2009; 36(3):472-477. <http://dx.doi.org/10.3899/jrheum.080732> PMID 19286859
5. Sled E, Sheehy L, Felson DT, Costigan, PA, Lam M, Cooke TDV, Reliability of lower limb alignment measures using an established landmark-based method with customized computer software program. Rheumatol Int 2011; 31(1): 71-77. <http://dx.doi.org/10.1007/s00296-009-1236-5> PMID 19882339 PMCID PMC3048894.



Appendix A. Comparison of Project 60 and Project 32 Measurements

A.1 Agreement between HKA measurements

In general, the exact same participants had HKA measured in both projects, but there are a number of people only measured in one of the projects; there are about 200 people only measured in Project 32 at the 12-month visit and about 50 people only measured in Project 32 at the 48-month visit.

Agreement between readings from both projects is good, with an ICC of 0.98 at any visit. A Bland-Altman analysis (Figure A.1) shows a mean difference of 0.1 degrees between Project 60 and Project 32, with 95% limits of agreement of ± 1.2 degrees.

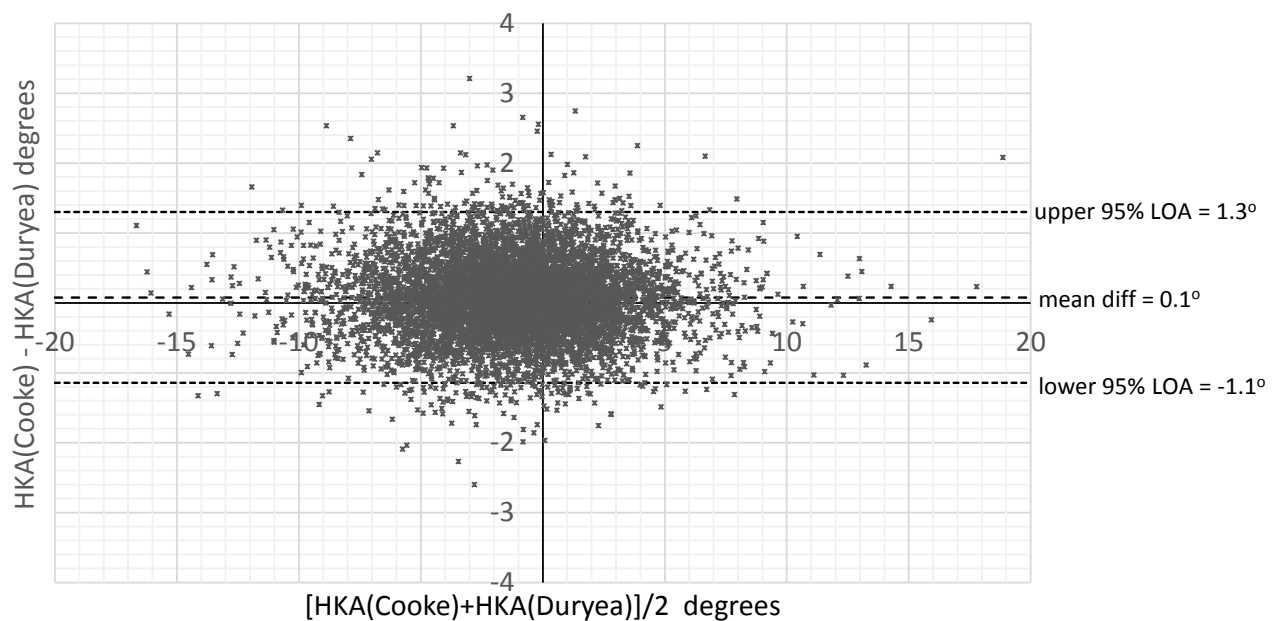


Figure A.1. Bland-Altman Plot showing agreement between HKA measured in Project 60 HKA(Cooke), and Project 32 HKA(Duryea) showing the mean difference along with the 95% limits of agreement (LOA).

A.2 Agreement on category of malalignment

If varus malalignment is defined as $HKA \leq -2$ degrees, valgus malalignment as $HKA \geq +2$ degrees, and neutral alignment as anything in between; then the agreement from comparing Project 60 and Project 32 on presence and direction of malalignment is good (weighted kappa=0.88 [0.87-0.89]) with 91% of limbs classified the same in both projects.